Affine Transform

By,

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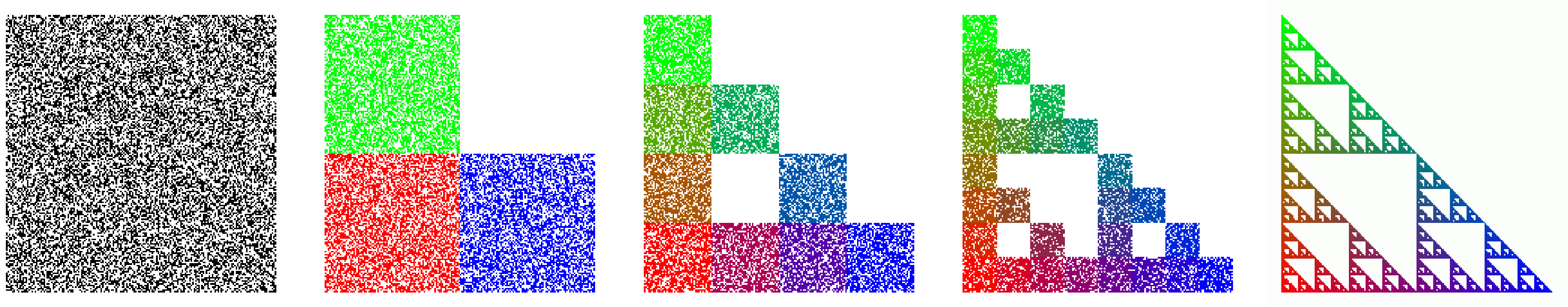
What is a Transformation?

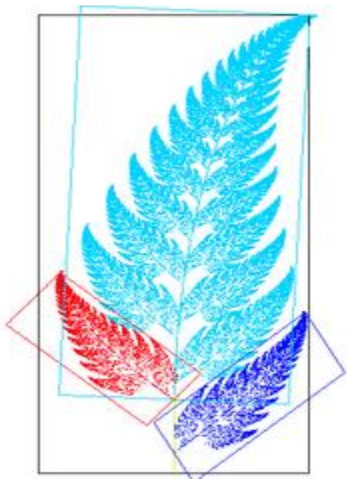
- Maps points (x, y) in one coordinate system to points (x', y') in another coordinate system

$$x' = ax + by + c$$

$$y' = dx + ey + f$$

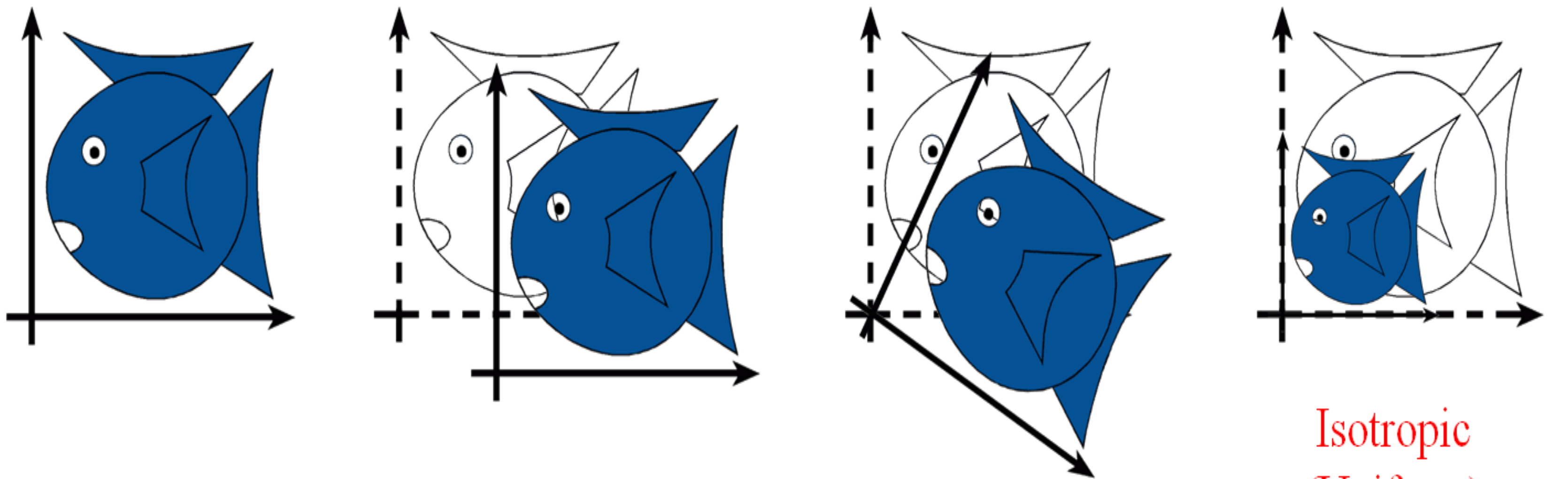
- For example, IFS:





An image of a **fern-like fractal** (Barnsley's fern) that exhibits affine self-similarity.
Each of the leaves of the fern is related to each other leaf by an affine transformation.

Simple Transformations



Isotropic Scaling having a physical property which has the same value when measured in different directions.

Transformations are used:

- ❖ Position objects in a scene (modeling)
- ❖ Change the shape of objects
- ❖ Create multiple copies of objects
- ❖ Projection for virtual cameras
- ❖ Animations

Affine Transform

- **Affine Transformation Describe** –
- How points, lines and planes are allowed to be transformed.
- Affine transformation is a linear mapping method that preserves points, straight lines, and planes.
- Sets of parallel lines remain parallel after an affine transformation.
- The affine transformation technique is typically used to correct for geometric distortions or deformations that occur with non-ideal camera angles.
- For example, satellite imagery uses affine transformations to correct for wide angle lens distortion, panorama stitching, and image registration.

Affine Transform

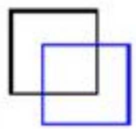
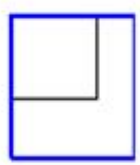
- The transformation that maps the pixel at the coordinates(x,y) to a new coordinate position is given as a pair of transformation equations.
- In this transform, straight lines are preserved and parallel lines remain unchanged.
- It is described mathematically as

$$x' = T_x (x,y)$$
$$y' = T_y (x,y)$$

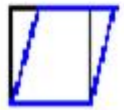
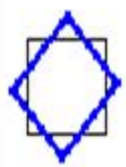
T_x and T_y are expressed as **Polynomials**. The linear equation gives an Affine Transform.

$$x' = a_0 x + a_1 y + a_2$$
$$y' = b_0 + b_1 y + b_2$$

Affine Transform

Affine Transform	Example	Transformation Matrix	
Translation		$\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$	t_x specifies the displacement along the x axis t_y specifies the displacement along the y axis.
Scale		$\begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$	s_x specifies the scale factor along the x axis s_y specifies the scale factor along the y axis.

Affine Transform

Shear		$\begin{bmatrix} 1 & sh_x & 0 \\ 1 & sh_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$	sh_x specifies the shear factor along the x axis sh_y specifies the shear factor along the y axis.
Rotation		$\begin{bmatrix} \cos(q) & \sin(q) & 0 \\ -\sin(q) & \cos(q) & 0 \\ 0 & 0 & 1 \end{bmatrix}$	q specifies the angle of rotation.

Affine Transform

It is expressed in the Matrix form as

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{pmatrix} a_0 & a_1 & a_2 \\ b_0 & b_1 & b_2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

The affine transform is a compact way of representing all transformations. The given equation represents all transformations.

Translation is the situation where $a_0 = 1$, $a_1 = 0$,and $a_2 = t_{x,}$ $b_0 = 0$, $b_1 = 1$ and $b_2 = t_{y.}$

Scaling Transformation-$a_0 = Sx$, $b_1 = Sy$ and $a_1 = 0$, $a_2 = 0$, $b_0 = 0$, $b_2 = 0$;

Rotation is a situation where $a_0 = \text{Cos}\Theta$, $a_1 = -\sin\Theta$, $b_0 = \sin\Theta$,

$b_1 = \text{Cos}\Theta$, $a_2 = 0$, $b_2 = 0$;

Horizontal shear is performed when $a_0 = 1$, $b_0 = 1$, $a_1 = Sh_x$, $a_2 = 0$, $b_1 = Sh_y$ and $b_2 = 0$.

Affine Transform Example

Affine Transformation Example:

1 Consider an image point [2,2]. Perform the following operations and show the results of these transforms.

(a) Translate the image right by 5 units

(b) Perform a scaling operation in both x-axis and y-axis by 4 units.

(c) Rotate the image in x-axis by 45°.

(d) Perform horizontal skewing by 45°

(e) Perform mirroring about x-axis.

Soln:- Translation of the image right by 5 units means that

$$t_x = 5, \quad t_y = 0.$$

So the translation matrix is given as

$$T = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Therefore,

$$x' = T \times x$$

$$= \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & 2 & 1 \end{bmatrix}^T$$

$$= \begin{bmatrix} 7 & 2 & 1 \end{bmatrix}^T$$

Affine Transform Example

(b) Scaling by 4 units in bothe the directions means that

$$S = \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times [2, 2, 1]$$

$$= [8 \; 8 \; 1]^T$$

Affine Transform Example

(c) Rotating the image in x-axis by 45°

$$R = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos 45 & -\sin 45 & 0 \\ \sin 45 & \cos 45 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.707 & -0.707 & 0 \\ 0.707 & 0.707 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & 2 & 1 \end{bmatrix}^T$$

$$= \begin{bmatrix} 0 & 0.2828 & 1 \end{bmatrix}^T$$

The fraction 0.2828 is rounded to 3

$$\text{Result} = \begin{bmatrix} 0 & 3 & 1 \end{bmatrix}^T$$

Affine Transform Example

(d) Performing horizontal skewing by 45°

$$\text{skew} = \begin{bmatrix} 1 & 0 & 0 \\ \tan\phi & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \tan 45 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & 2 & 1 \end{bmatrix}^T$$

$$= \begin{bmatrix} 2, & 4 & 1 \end{bmatrix}^T$$

Affine Transform Example

(e) Performing mirroring about x-axis

$$mx = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{mirroring} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & 2 & 1 \end{bmatrix}$$

$$\text{Result} = \begin{bmatrix} 2 & -2 & 1 \end{bmatrix}^T$$

Thanks